

Your Code

```
1 import java.util.Scanner;
2
3 public class SmartDocumentEditor {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter a paragraph:");
8         String p= sc.nextLine();
9
10        StringBuffer his= new StringBuffer();
11        his.append("Original Document:\n").append(p).append("\n\n");
12
13        int chars = p.length();
14        String[] wordsarr = p.trim().split("\\s+");
15        int words = wordsarr.length;
16        String[] sentarr = p.split("[.!?]");
17        int sent = 0;
18        for (String s : sentarr) {
19            if (!s.trim().isEmpty())
20                sent++;
21        }
22
23        his.append("Character Count: ").append(chars).append("\n");
24        his.append("Word Count: ").append(words).append("\n");
25        his.append("Sentence Count: ").append(sent).append("\n\n");
26
27        System.out.println("Enter the word to replace:");
28        String oldWord = sc.nextLine();
29        System.out.println("Enter the new word:");
30        String newWord = sc.nextLine();
31
32        String rP = p.replaceAll("\\b" + oldWord + "\\b", newWord);
33        his.append("After Replacing \"").append(oldWord)
34            .append("\" with \"").append(newWord).append("\":\n")
35            .append(rP).append("\n\n");
36
37        String[] sentToRev= rP.split("(?<=[.!?])\\s*");
38        StringBuilder revSent = new StringBuilder();
39
40        for (String sentence : sentToRev) {
41            if (!sentence.trim().isEmpty()) {
42                StringBuilder sb = new StringBuilder(sentence.trim());
43                revSent.append(sb.reverse()).append(" ");
44            }
45        }
46
47        his.append("After Reversing Each Sentence:\n")
48            .append(revSent.toString().trim()).append("\n\n");
49
50        StringBuilder fD = new StringBuilder();
51        fD.append("=== FINAL FORMATTED DOCUMENT ===\n");
52        fD.append("Total Characters: ").append(chars).append("\n");
53        fD.append("Total Words: ").append(words).append("\n");
54        fD.append("Total Sentences: ").append(sent).append("\n");
55        fD.append("Modified & Reversed Content:\n");
56        fD.append(revSent.toString().trim());
57
58        System.out.println("\n----- ORIGINAL DOCUMENT -----");
59        System.out.println(p);
60
61        System.out.println("\n----- EDITING HISTORY -----");
62        System.out.println(his.toString());
63
64        System.out.println("\n----- FINAL DOCUMENT -----");
65        System.out.println(fD.toString());
66
67        sc.close();
68    }
69 }
```

Input

Python is a programming language.
Python is powerful and versatile.
Python
Java

Output

```
Enter a paragraph:
Enter the word to replace:
Enter the new word:

----- ORIGINAL DOCUMENT -----
Python is a programming language
```

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc=new Scanner(System.in);
5         ArrayList<Patient> list = new ArrayList<>();
6         int n=sc.nextInt();
7         sc.nextLine();
8         for(int i=0;i<n;i++){
9             int id=sc.nextInt();
10            sc.nextLine();
11            String name=sc.nextLine();
12            String dept=sc.nextLine();
13            list.add(new Patient(id,name,dept));
14        }
15        System.out.println("All Patients: ");
16        for(Patient p: list){
17            System.out.println(p.id+ " "+p.name+" "+p.dept);
18        }
19        int id=sc.nextInt();
20        System.out.println("The Entered patient ID to update is : "+id);
21        sc.nextLine();
22        String newName=sc.nextLine();
23        System.out.println("The Entered name to update : "+newName);
24        for (Patient p:list){
25            if(p.id==id){
26                p.name=newName;
27            }
28        }
29        String dep=sc.nextLine();
30        System.out.println("Enter Department to search : "+dep);
31        System.out.println("Patients in "+dep+" Department: ");
32        for(Patient p:list){
33            if(p.dept.equalsIgnoreCase(dep)){
34                System.out.println(p.id+" "+p.name);
35            }
36        }
37        System.out.println("\nFirst Registerd Patient: ");
38        System.out.println(list.get(0).id+" "+list.get(0).name);
39        System.out.println("Last Registered Patient: ");
40        System.out.println(list.get(list.size()-1).id+" "+list.get(list
41    }
42 }
43
44 class Patient {
45     int id;
46     String name;
47     String dept;
48
49     Patient(int id, String name, String dept) {
50         this.id = id;
51         this.name = name;
52         this.dept = dept;
53     }
54 }
```

Input

```
3
101
ram
Cardiology
```

Output

```
All Patients:
101 ram Cardiology
102 ravi Neurology
103 Kumar Cardiollogy
The Entered patient ID to update
is : 102
```

Your Code

```
1 import java.util.*;
2
3 public class AirportQueue {
4     public static void main(String[] args) {
5         Scanner sc= new Scanner(System.in);
6         Queue<String> queue = new LinkedList<>();
7         PriorityQueue<String> vip = new PriorityQueue<>();
8         int n=sc.nextInt();
9         sc.nextLine();
10        for (int i=0;i<n;i++){
11            queue.add(sc.nextLine());
12        }
13        System.out.println("Current Queue: "+queue);
14        System.out.println("Next Passenger: "+queue.peek());
15        System.out.println("Boarded Passenger: "+queue.poll());
16        System.out.println("Remaining Queue: "+queue);
17        int m= sc.nextInt();
18        sc.nextLine();
19        for (int i=0;i<m;i++){
20            vip.add(sc.nextLine());
21        }
22        System.out.println("VIP Queue: "+vip);
23    }
24 }
25 }
```

Input

3
Ram
Ravi
Kumar

Output

Current Queue: [Ram, Ravi, Kumar]
Next Passenger: Ram
Boarded Passenger: Ram
Remaining Queue: [Ravi, Kumar]
VIP Queue: [Arun, Bala]

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String args[]){
4         Scanner sc= new Scanner(System.in);
5         LinkedHashSet<Integer> set=new LinkedHashSet<>();
6         TreeSet<Integer> s=new TreeSet<>();
7
8         int n=sc.nextInt();
9         for (int i=0;i<n;i++){
10             int id=sc.nextInt();
11             set.add(id);
12             s.add(id);
13         }
14         System.out.println("Certificates (Insertion Order): "+ set);
15         System.out.println("Certificate (Sorted Order): "+s);
16         System.out.println("Total Unique Certificates: "+set.size());
17         int search=sc.nextInt();
18         System.out.println("The Entered Certificate Id to search is: "+search);
19         if(set.contains(search)){
20             System.out.println("Certificate Found");
21         }else{
22             System.out.println("Certificate Not Found");
23         }
24     }
25 }
26 }
27 }
```

Input

4
100
99
102

Output

Certificates (Insertion Order):
[100, 99, 102, 104]
Certificate (Sorted Order): [99,
100, 102, 104]
Total Unique Certificates: 4
The Entered Certificate Id to

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String args[]){
4         Scanner sc= new Scanner(System.in);
5         HashMap<Integer,Product> map=new HashMap<>();
6         int n=sc.nextInt();
7         for(int i=0;i<n;i++){
8             int id = sc.nextInt();
9             sc.nextLine();
10            String name=sc.nextLine();
11            String category=sc.nextLine();
12            int quantity=sc.nextInt();
13            double price=sc.nextDouble();
14            map.put(id,new Product(id,name,category,quantity,price));
15        }
16        int sid=sc.nextInt();
17        if(map.containsKey(sid)){
18            Product p=map.get(sid);
19            System.out.println("Product Found");
20            System.out.println(p.id+" "+p.name+" "+p.category+" "+p.quantity);
21        }else{
22            System.out.println("Product Not Found");
23        }
24        double total=0;
25        for (Product p: map.values()){
26            total+=p.quantity*p.price;
27        }
28        System.out.println("Total Inventory Value = "+total);
29        sc.nextLine();
30        String cat=sc.nextLine();
31        for(Product p: map.values()){
32            if(p.category.equalsIgnoreCase(cat)){
33                System.out.println(p.id+" "+p.name+" "+p.quantity+" "+p.price);
34            }
35        }
36        for(Product p: map.values()){
37            System.out.println(p.id+" "+p.name+" "+p.category+" "+p.quantity);
38        }
39    }
40 }
41 class Product{
42     int id;
43     String name,category;
44     int quantity;
45     double price;
46     Product(int id,String name,String category,int quantity,double price){
47         this.id=id;
48         this.name=name;
49         this.category=category;
50         this.quantity=quantity;
51         this.price=price;
52     }
53 }
```

Input

```
3
101
laptop
Electronics
```

Output

```
Product Found
101 laptop Electronics 5 50000.0
Total Inventory Value = 263000.0
101 laptop 5 50000.0
102 mouse 20 500.0
101 laptop Electronics 5 50000.0
```